# 1 Matrix of an Isomorphism

Let $V$ and $W$ be finite-dimensional vector spaces and suppose $T : V \to W$ is an isomorphism. Then $\dim V = \dim W = n$, so $[T] = [T^{-1}]$ are $n \times n$ matrices. Since $T^{-1}T = \mathrm{id}_V$ and $TT^{-1} = \mathrm{id}_W$ and $[\mathrm{id}_V] = [\mathrm{id}_W] = I$ (the $n \times n$ identity matrix), we have

$$[T^{-1}T] = [T^{-1}][T] = I, [TT^{-1}] \qquad = [T][T^{-1}] = I.$$

**Definition 1.1 (Invertible Matrix)**

Let $A$ be an $n \times n$ matrix. If there exists and $n \times n$ matrix $B$ such that $AB = BA = I$, then $A$ is said to be invertible or non-signular and $B$ is said to be the inverse of $A$, which we denote by $B = A^{-1}$. A matrix which is not invertible is called *non-invertible* or *singular*.

The matrix of an isomorphism of finite-dimensional vector spaces is therefore an invertible matrix.

**Exercise 1.2**

Prove that the inverse of a matrix $A$, if it exists, is unique.

**Proof:**

Pass the matrix $A$ to its linear map $T$. An inverse matrix corresponds to an inverse of the map, so it suffices to prove inverses are unique. But, if we have an inverse $H_1$ and an inverse $H_2$, then:

$$H_1 = H_1 I = H_1(TH_2) = (H_1T)H_2 = IH_2 = H_2,$$

so $H_1 = H_2$, and therefore their matrices are identical

**Exercise 1.3**

Prove that if $A$ and $B$ are invertible $n \times n$ matrices, then $AB$ is an invertible $n \times n$ matrix with $(AB)^{-1} = B^{-1}A^{-1}$. Similarly, any finite product $A_1 \cdot \ldots \cdot A^k$ of invertible $n \times n$ matrices is invertible with $(A_1 \ldots A_k)^{-1} = A_k^{-1} \ldots A_1^{-1}$.

**Proof:**

It suffices to show this for the $n = 2$ case by induction. Then, we simply see $B^{-1}A^{-1}$ is an inverse since $AB(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$.

# 2 Elementary Matrices

**Definition 2.1 (Elementary matrix)**

An $n \times n$ matrix is said to be an *elementary matrix* if it can be obtained from the $n \times n$ identity matrix by a single elementary row operation.

**Example 2.2**

Let $c$ be a constant. Verify that the complete list of $2 \times 2$ elementary matrices is

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}, \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} (c \neq 0), \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix} (c \neq 0).$$



**Proposition 2.3**

If $e : M^{n\times n}(F) \to M^{n\times n}(F)$ is an elementary row operation and $E$ the $n \times n$ elementary matrix $E = e(I)$, then for every $n \times n$ matrix $A$, $e(A) = EA$.



**Proof:**

We consider each type of elementary row operation separately.

1. Suppose $r \neq s$ and let $e$ be the elementary row operation which replaces row $r$ by row $r$ plus $c$ times row $s$. We need to show that $EA = e(A)$. We can write the elements of corresponding elementary matrix $E$ as

$$E_{ik} = \Big\{ \delta_{ik} i \neq r \delta_{rk} + c\delta_{sk} i = r$$

Then, if $i \neq r$,

$$(EA)_{ij} = \sum_{k=1}^{m} E_{ik}A_{kj} = \sum_{k=1}^{m} \delta_{ik}A_{kj} = A_{ij}$$

and for $i = r$,

$$(EA)_{rj} = \sum_{k=1}^{m} E_{rk}A_{kj} = \sum_{k=1}^{m} (\delta_{rk} + c\delta_{sk})A_{kj} = A_{rj} + cA_{sj},$$

which shows $EA = e(A)$ as desired.

2. Suppose $r \neq s$ and let $e$ be the elementary row operation which exchanges rows $r$ and $s$. the elements of the corresponding elementary matrix are given by

$$\Big\{ \delta_{ik} i \neq r, s\delta_{rk}, i = s\delta_{sk}, i = r.$$

Then, if $i \neq r$,$s$,

$$(EA)_{ij} = \sum_{k=1}^{m} E_{ik}A_{kj} = \sum_{k=1}^{m} \delta_{ik}A_{kj} = A_{ij}$$

if $i = r$,

$$(EA)_{rj} = \sum_{k=1}^{m} E_{rk}A_{kj} = \sum_{k=1}^{m} \delta_{sk}A_{kj} = A_{sj}$$

and if $i = s$,

$$(EA)_{sj} = \sum_{k=1}^{m} E_{sk}A_{kj} = \sum_{k=1}^{m} \delta_{rk}A_{kj} = A_{rj},$$

which shows $EA = e(A)$.

**5.** Let $e$ be the elementary row operation which multiplies rows $r$ by $c$, $c \neq 0$. We need to show that $EA = e(A)$. The corresponding elementary matrix can be written as

$$E_{ik} = \Big\{ \delta_{ik} i \neq rc\delta_{rk}, i = r.$$

Then, if $i \neq r$,

$$(EA)_{ij} = \sum_{k=1}^{m} E_{ik} A_{kj} = \sum_{k=1}^{m} \delta_{ik} A_{kj} = A_{ij},$$

if $i \neq r$,

$$(EA)_{rj} = \sum_{k=1}^{m} E_{rk} A_{kj} = \sum_{k=1}^{m} (c\delta_{rk}) A_{kj} = cA_{rj}$$

which shows $EA = e(A)$.





**Exercise 2.4**

**10.**

1. Let $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ be the elementary matrix obtained from the $2 \times 2$ identity matrix by replacing $R2 \to R2 - 3R_1$. Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Show that the matrix product $EA$ is equal to the result of performing the row replacement $R2 \to R2 - 3R1$ on $A$.
2. Let $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ be the elementary matrix obtained from the $2 \times 2$ identity matrix b the interchange $R_1 \iff R_2$. Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ . show that the matrix product $EA$ is equal to the result of interchanging the two rows of $A$.



**Solution 1**

1. We have:

$$EA = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix},$$

exactly $R2 \to R2 - 3R1$.

2. We have:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix},$$

exactly the interchange $R1 \longleftrightarrow R2$



**Corollary 2.5**

Let $A$ and $B$ be $m \times n$ matrices. Then $B$ is row-equivalent to $A$ if and only if $B = PA$, where $P$ is a product of $m \times m$ elementary matrices.

**Proof:**

Suppose $B$ is row-equivalent to $A$. Let $E_1, E_2, ..., E_s$ be the elementary matrices corresponding to some sequence of elementary row operations which carries $A$ into $B$. Then $B = (E_s...E_1)A$. Suppose $B = PA$ where $P = E_s...E_2E_1$ and the and the $E_i$ are $m \times m$ elementary matrices. Then $E_1A$ is row-equivalent to $A$ and $E_2(E_1A)$ is row-equivalent to $A$, so transitivity gives $E_1E_1A$ is row-equivlaent to $A$, and the result follows by induction.

**Proposition 2.6**

An elementary matrix $E$ is invertible. The inverse of $E$ is the elementary matrix of the same type that transforms $E$ back into $I$.

**Proof:**

Let $E$ be the elementary matrix corresponding to the elementary row operation $e$. If $e'$ is the inverse operation of $e$ and $E' = e'(I)$, then $EE' = e(E') = e(E'(I)) = I$, so $E$ is invertible and $E' = E^{-1}$.

**Example 2.7**

1. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
2. $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -c \\ 0 & 1 \end{bmatrix}$.
3. $\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -c & 1 \end{bmatrix}$
4. When $c \neq 0$, $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{c} & 0 \\ 0 & 1 \end{bmatrix}$ ,, $\begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}^1 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{c} \end{bmatrix}$

# 3 Invertible Matrix Theorem

**Theorem 3.1 (Invertible Matrix Theorem)**

If $A$ is an $n \times n$ matrix the following are equivalent.

1. $A$ is invertible.
2. $A$ is row-equivalent to the $n \times n$ identity matrix
3. $A$ is a product of elementary matrices.

**Proof:**

We will show that $(i) \to (ii) \to (iii) \to (i)$. Assume $(i)$. Let $R$ be the RRRF of $A$. Then, $R = E_k...E_2E_1A$ where $E_1, ..., E_k$ are elementary. Since these are all invertible, we see $A = E_1^{-1}...E_k^{-1}R$. Since products of invertible matrices are invertible, we see $A$ is invertible if and only if $R$ is invertible. Since $R$ is a square matrix in RREf, $R$ is invertible if and only if it is the $n \times n$ identity matrix (otherwise, it would have a row of zeroes and any such matrix is singular).Hence, $(i) \to (ii)$. $(iii)$

follows immediately from $(ii)$. $(i)$ follows immediately from the third by the exercise.

**Corollary 3.2 (How to compute $A^{-1}$)**

If $A$ is an invertible $n \times n$ matrix and if a sequence of elementary row operations reduces $A$ to the identity, then that same sequence of operations when applied to $I$ yields $A^{-1}$

**Proof:**

If $E_k...E_1A = I$ where each $E_j$ is an elementary matrix, then multiplying both sides on the right by $A^{-1}$ gives $E_k...E_1 = A^{-1}$.

Corollary 3.2 gives an algorithm to compute $A^{-1}$: we simply row reduce the augmented matrix $[A \mid I]$. If $A$ is row-equivalent to $I$, $[A \mid I]$ is row equivalent to $[I \mid A^{-1}]$. Otherwise, $A$ does not have an inverse.

**Example 3.3**

We find the inverse of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ by performing Gauss-Jordan elimination on the augmented matrix

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right] \; whichgives \; \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array}\right]$$

so $A^{-1} \begin{bmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & 4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{bmatrix}$.

**Exercise 3.4**

Find the inverse of the matrix $A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 0 & 2 \\ -1 & -2 & 4 \end{bmatrix}$.

**Solution**

Augment with the identity:

$$\left[\begin{array}{ccc|ccc} 3 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ -1 & -2 & 4 & 0 & 0 & 1 \end{array}\right]$$

**Exercise 3.5**

Prove that a linear systems $Ax = y$ of $n$ equations in $n$ unknowns has a unique solution if and only if $A$ is invertible.



**Proof:**

Pass from the matrix perspective to the linear maps perspective and take the contrapositive. If $A$ is not invertible, $A$ is not an isomorphism (since isomorphism are invertible), and therefore is not injective or surjective (which implies it is not the other by the fundamental theorem of linear maps). Then, either it does not have a solution (since the map is not surjective), and if it has a solution $x_0$, $x_0 + v$ for $v \in \ker(A)$ gives the same output $y$, so the solution is not unique.





**Exercise 3.6**

Consider the linear systems

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 5x_2 + 3x_3 &= 3 \\ x_1 + 8x_3 &= 17 \end{aligned}$$

The coefficient matrix of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$, which has inverse $A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$. Use $A^{-1}$ to find the unique solution of the system.



**Solution**

This system can be rewritten as:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix},$$

so multiplying by $A^{-1}$ gives:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$